

Disc caliper - design brief

AI BRAKING / AB-DC / C0 / 2026-09-07

A fixed opposed-pad bridge concept around the common 400 mm disc. A purchased spring-applied hydraulic-release cartridge is represented by its envelope; its internal mechanism is not designed here.

TARGETS (not validated)

Disc diameter / thickness: 400 / 20 mm. Shared geometry

Pad planform: 70 × 60 mm. Concept area before corner / retention detail

Study torque: 1,000 N·m. Unverified requirements hypothesis

Effective radius: 155 mm. Uniform-pressure rectangular-pad approximation

Normal force per face: 10.75 kN. $T / (2\mu r)$, $\mu = 0.30$

DESIGN DECISIONS

A fixed bridge establishes a direct load path; removable pad retainers are a serviceability hypothesis.

Use a bought release cartridge for the first test rig rather than developing seals and spring stacks simultaneously.

Share the rotor/hub pattern with the positive-actuated brake study.

CANDIDATE MATERIALS

Bridge and cheeks: S355J2+N plate for prototype; machined critical faces after fabrication.

Pad backing: C45 steel candidate, with purchased qualified lining.

Tie fasteners: purchased certified fasteners; grade, preload and locking method unresolved.

Assembly and engineering gates

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ASSEMBLY INTENT

- 1. Locate bridge and cheek plates with interface holes aligned.
- 2. Install replaceable backing plates, lining and retainer pins.
- 3. Mount supplier-approved actuator cartridges on the outer interfaces.
- 4. Centre rotor in the cold gap; verify hot-growth allowance analytically.
- 5. Approve joint preload, retainer security and release stroke before guarded testing.

OPEN DESIGN RISKS

- 1. Bridge deflection, pad pressure distribution and hot clearance are not verified.
- 2. Cartridge load path and equalization are interface requirements, not functional geometry.
- 3. Pressure, spring force and release stroke must come from the selected supplier.

VALIDATION REQUIRED

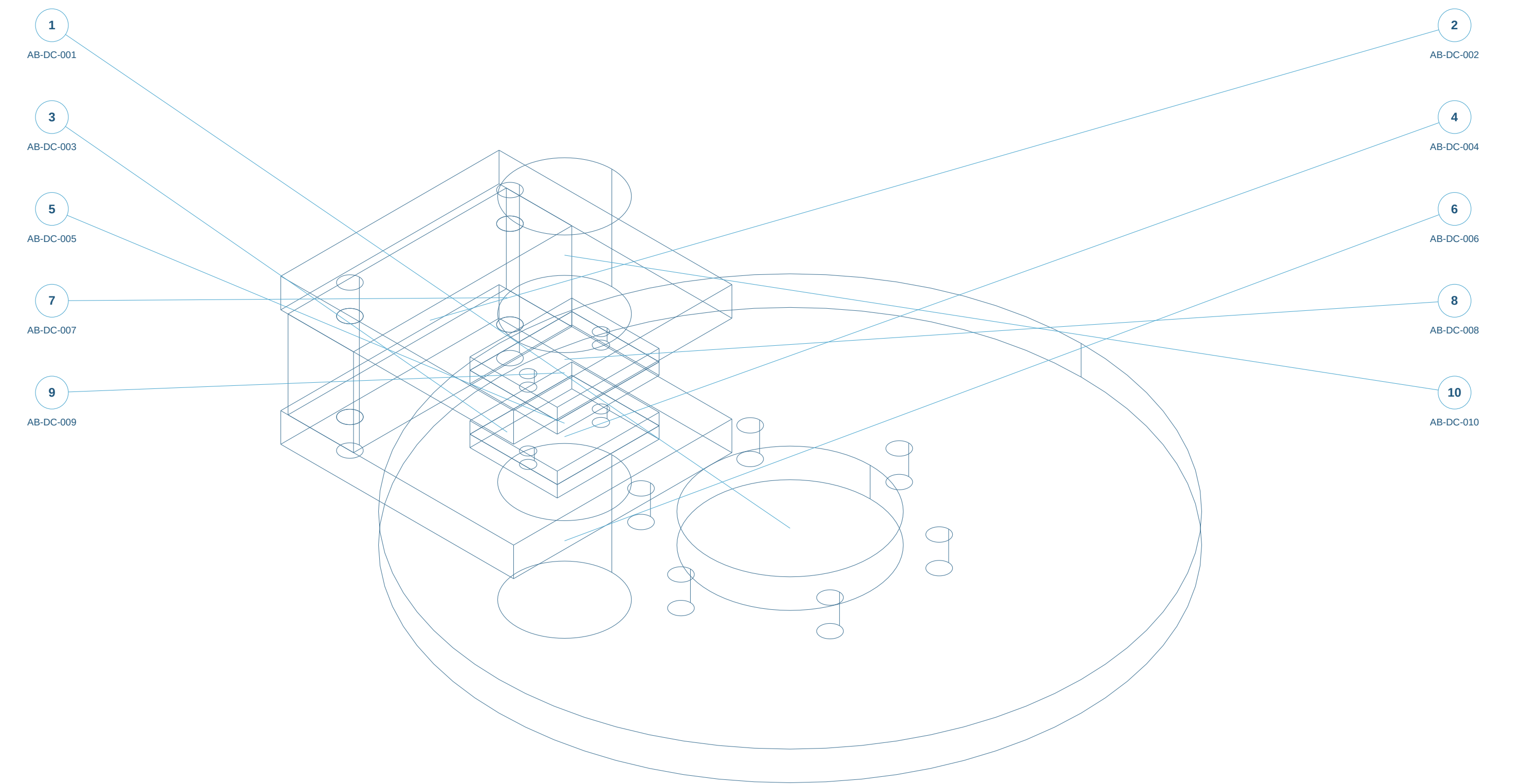
- 1. Contact/nonlinear bridge FEA with supplier actuator loads.
- 2. Thermal distortion, wear travel and worst-case tolerance stack.
- 3. Torque versus pressure/release state, pad change, leakage and endurance tests.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

- 1. Advisory position / wear sensing with replaceable brackets; no AI-controlled brake release.
- 2. Record supplier sensor range, supply, connector, EMC, ingress and temperature limits before selection.
- 3. Define independent machine safety interface and diagnostic fault response with the responsible controls engineer.

Assembly reference / numbered components

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Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

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AB-DC-001 | Common disc reference | shared-concept

C45 candidate; Saw/laser blank; CNC critical faces and bores

AB-DC-002 | Bridge beam | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-DC-003 | Cheek plate -1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-DC-004 | Pad backing -1 | make-concept

C45 candidate; Saw/laser blank; CNC critical faces and bores

AB-DC-005 | Friction pad -1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Purchase matched compound and approved bonding/retention

AB-DC-006 | Spring-release cartridge -1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Purchase cartridge; mounting interface unassigned

AB-DC-007 | Cheek plate 1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-DC-008 | Pad backing 1 | make-concept

C45 candidate; Saw/laser blank; CNC critical faces and bores

AB-DC-009 | Friction pad 1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Purchase matched compound and approved bonding/retention

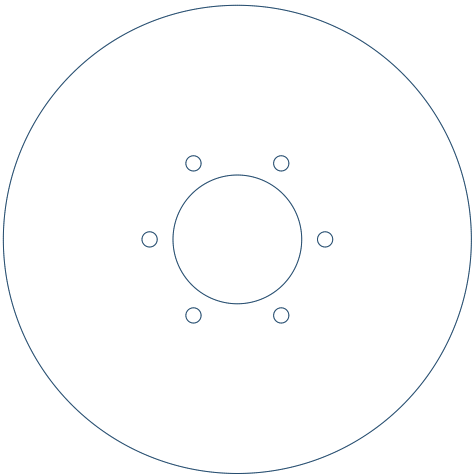
AB-DC-010 | Spring-release cartridge 1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Purchase cartridge; mounting interface unassigned

AB-DC-001 / Common disc reference

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TOP X-Y



FRONT X-Z



C45 candidate

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 400.00 / 400.00 / 20.00 mm

Common AB-ROT disc geometry; see rotor package for hub.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



AB-DC-002 / Bridge beam

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TOP X-Y



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 150.00 / 45.00 / 60.00 mm

Bridge width 150; depth 45; axial span 60.

Tie interface bores are concept only; structural joint not released.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

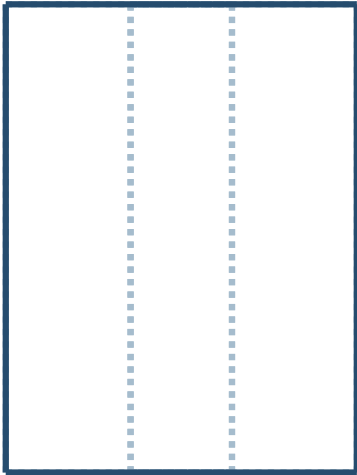
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



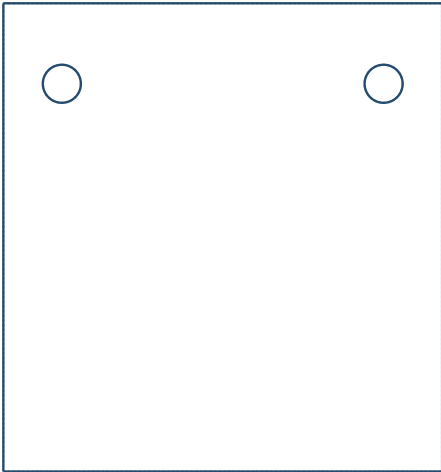
SIDE Y-Z



AB-DC-003 / Cheek plate -1

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 150.00 / 160.00 / 20.00 mm

150 x 160 x 20; mounting holes local X +/-55, Y 52.5.

Faces mate bridge at Z +/-30; preload/fastener detail pending.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

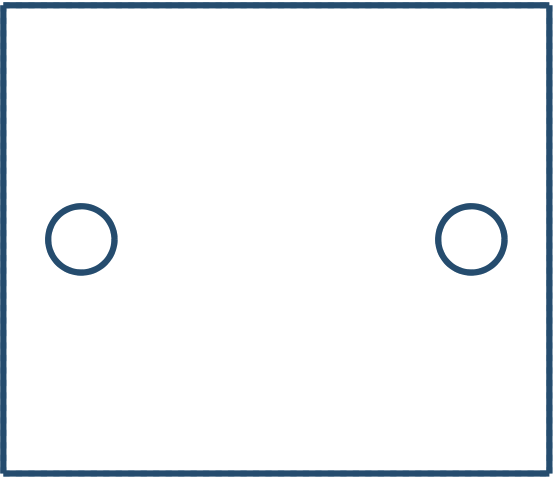
SIDE Y-Z



AB-DC-004 / Pad backing -1

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TOP X-Y



FRONT X-Z



C45 candidate

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 70.00 / 60.00 / 8.00 mm

70 x 60 x 8; two diameter 8.5 retainer holes.

Retention detail must avoid lining damage.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

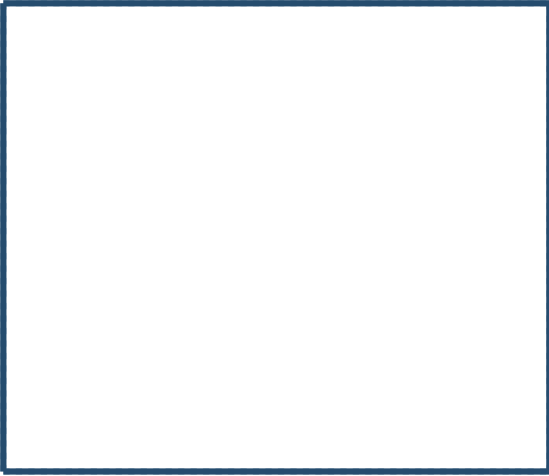
SIDE Y-Z



AB-DC-005 / Friction pad -1

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Purchase matched compound and approved bonding/retention

Overall X/Y/Z bounds: 70.00 / 60.00 / 8.00 mm

Pad inner face Z +/-11; 1 mm cold study clearance to disc.

Uniform pressure effective radius approximated at 155 mm.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

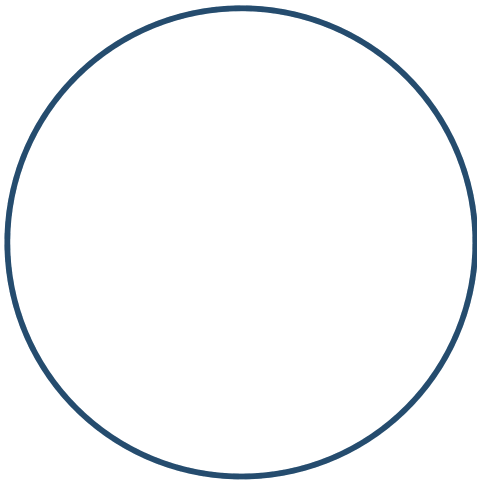
SIDE Y-Z



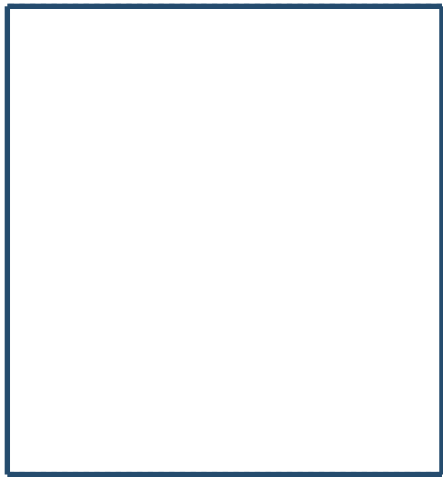
AB-DC-006 / Spring-release cartridge -1

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Purchase cartridge; mounting interface unassigned

Overall X/Y/Z bounds: 65.00 / 65.00 / 70.00 mm

Diameter 65 x 70 envelope only.

No pressure port, piston, seals or spring mechanism modelled.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

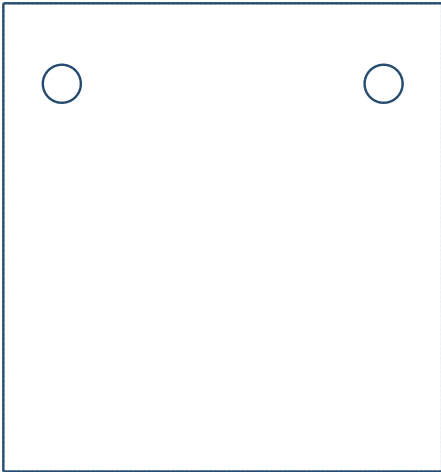
SIDE Y-Z



AB-DC-007 / Cheek plate 1

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 150.00 / 160.00 / 20.00 mm

150 x 160 x 20; mounting holes local X +/-55, Y 52.5.

Faces mate bridge at Z +/-30; preload/fastener detail pending.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

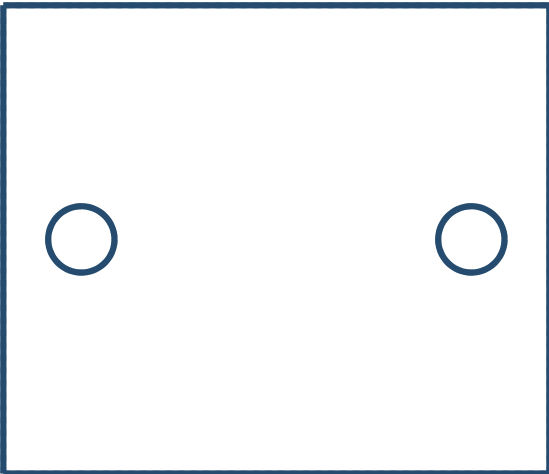
SIDE Y-Z



AB-DC-008 / Pad backing 1

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TOP X-Y



FRONT X-Z



C45 candidate

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 70.00 / 60.00 / 8.00 mm

70 x 60 x 8; two diameter 8.5 retainer holes.

Retention detail must avoid lining damage.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

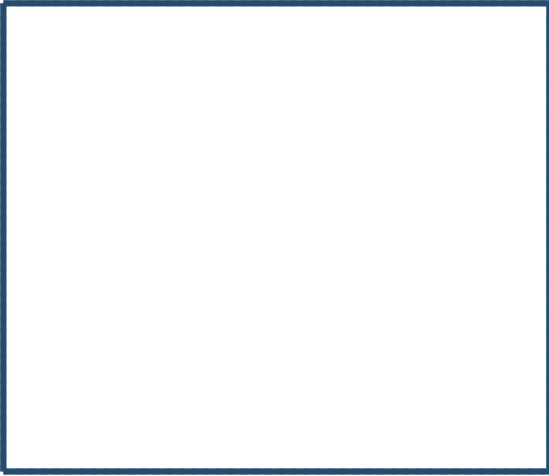
SIDE Y-Z



AB-DC-009 / Friction pad 1

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Purchase matched compound and approved bonding/retention

Overall X/Y/Z bounds: 70.00 / 60.00 / 8.00 mm

Pad inner face Z +/-11; 1 mm cold study clearance to disc.

Uniform pressure effective radius approximated at 155 mm.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

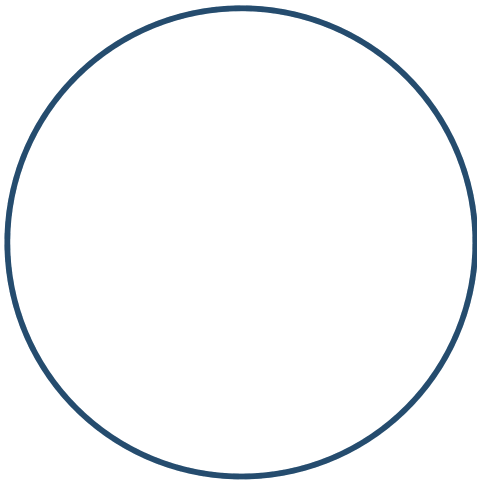
SIDE Y-Z



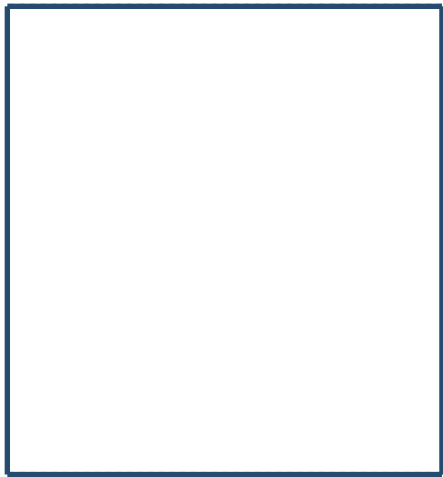
AB-DC-010 / Spring-release cartridge 1

AI BRAKING / AB-DC / C0 / 2026-09-07

TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Purchase cartridge; mounting interface unassigned

Overall X/Y/Z bounds: 65.00 / 65.00 / 70.00 mm

Diameter 65 x 70 envelope only.

No pressure port, piston, seals or spring mechanism modelled.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z

